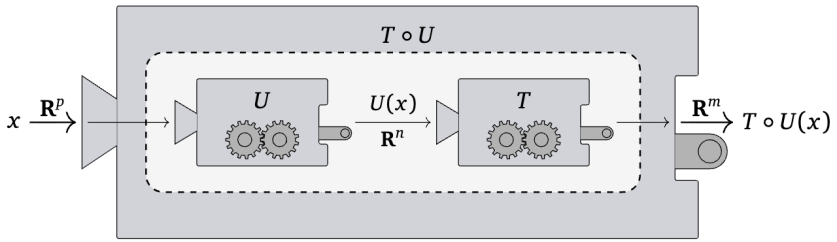
Matrix Multiplication

Composition of linear transformations

Composition means the same thing in linear algebra as it does in Calculus. Here is the definition.

Definition. Let $T: \mathbf{R}^n \to \mathbf{R}^m$ and $U: \mathbf{R}^p \to \mathbf{R}^n$ be transformations. Their ***composition*** is the transformation $T \circ U: \mathbf{R}^p \to \mathbf{R}^m$ defined by

$$(T \circ U)(x) = T(U(x)).$$



Domain and codomain of a composition.

- In order for $T \circ U$ to be defined, the codomain of U must equal the domain of T.
- The domain of $T \circ U$ is the domain of U.
- The codomain of $T \circ U$ is the codomain of T.

e.g. $T: \mathbb{R}^n \to \mathbb{R}^m$, $U: \mathbb{R}^p \to \mathbb{R}^n$, $T \circ U: \mathbb{R}^p \to \mathbb{R}^m$

Properties of composition. Let S, T, U be transformations and let c be a scalar. Suppose that $T: \mathbf{R}^n \to \mathbf{R}^m$, and that in each of the following identities, the domains and the codomains are compatible when necessary for the composition to be defined. The following properties are easily verified:

$$S \circ (T + U) = S \circ T + S \circ U \qquad (S + T) \circ U = S \circ U + T \circ U$$
$$c(T \circ U) = (cT) \circ U \qquad c(T \circ U) = T \circ (cU) \quad \text{if } T \text{ is linear}$$
$$T \circ \mathrm{Id}_{\mathbf{R}^n} = T \qquad \mathrm{Id}_{\mathbf{R}^m} \circ T = T$$
$$S \circ (T \circ U) = (S \circ T) \circ U$$

Composition of transformations is *not* commutative in general. That is, in general, $T \circ U \neq U \circ T$, even when both compositions are defined.

The sizes of the matrices in the matrix product.

- In order for AB to be defined, the number of rows of B has to equal the number of columns of A.
- The product of an $m \times n$ matrix and an $n \times p$ matrix is an $m \times p$ matrix.

Recipe: The row-column rule for matrix multiplication. Let A be an $m \times n$ matrix, let B be an $n \times p$ matrix, and let $C = AB$. Then the ij entry of C is the ith row of A times the jth column of B:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Here is a diagram:

$$\begin{pmatrix} a_{11} & \cdots & a_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ik} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mk} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1p} \\ \vdots & & \vdots & & \vdots \\ b_{k1} & \cdots & b_{kj} & \cdots & b_{kp} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{np} \end{pmatrix} = \begin{pmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1p} \\ \vdots & & \vdots & & \vdots \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{ip} \\ \vdots & & \vdots & & \vdots \\ c_{m1} & \cdots & c_{mj} & \cdots & c_{mp} \end{pmatrix}$$

ith row; jth column; ij entry

Matrix multiplication caveats.

- Matrix multiplication is not commutative: AB is not usually equal to BA, even when both products are defined and have the same size. See this example.
- Matrix multiplication does not satisfy the cancellation law: $AB = AC$ does not imply $B = C$, even when $A \neq 0$. For example,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix}.$$

- It is possible for $AB = 0$, even when $A \neq 0$ and $B \neq 0$. For example,

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Products and compositions.

The matrix of the composition of two linear transformations is the product of the matrices of the transformations.

$T(x) = Ax \quad U(x) = Bx$

$T \circ U(x) = T(U(x)) = T(Bx) = ABx$